

YASH GUPTA

SOFTWARE ENGINEER | eBPF | KERNEL SECURITY | DETECTION ENGINEERING | DATA SECURITY

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PROFILE

Software Engineer with 6 years of hands-on experience (Aug 2020 - Present) spanning cybersecurity, distributed systems, cloud infrastructure, and AI. Built kernel-level agents and detection pipelines across eBPF runtime tracing, process lineage, network interception, and sensitive-data egress detection. Experienced in turning low-level host and network telemetry into security context for threat detection, investigation, data lineage, DLP, and full data-egress observability.

Career note: Recently resuming active industry conversations after a brief health-related interruption.

EXPERIENCE

Founding Software Engineer | Matters.AI / OptIQ.AI

Feb 2024 - Present | Bengaluru, India

- Architected a Linux security agent and detection pipeline using eBPF runtime telemetry to observe process execution, socket activity, and data movement with low host overhead.
- Built process-lineage correlation across exec/fork activity, namespaces, containers, and network flows, giving detections the originating process and workload context needed for investigation.
- Developed network-interception and egress-observability capabilities across HTTP/2, SFTP, and additional outbound protocols, driving 90%+ protocol coverage for monitored traffic.
- Joined runtime and network telemetry with sensitive-data classification to detect risky egress and potential exfiltration paths, improving real-time threat visibility by 70%.
- Shipped enterprise security controls including SSO, RBAC, risk quantification, database connectors, access control, and data masking; production classification services operated at 99.9% uptime.

Independent Software Engineer | Self-Employed | Jan 2024 | Remote

Delivered short-form backend, cloud, and security engineering engagements for early-stage teams.

Software Engineer | Trilogy (Crossover)

Jan 2023 - Dec 2023 | Remote

- Built Terraform and CloudFormation analysis workflows, including a CLI and VS Code extension, to surface cloud configuration and security issues directly in developer workflows; caching cut average CLI latency by 60%.
- Designed the ingestion API and 6 AWS Step Functions for an eDiscovery processing system, preserving resource and event lineage through distributed stages while eliminating race conditions from two critical ingestion paths.
- Hardened detection-oriented processing by normalizing distributed pipeline outputs, improving end-to-end linting performance by 40%, and reducing LLM token usage by 35% through prompt optimization and deterministic parsing.

Software Engineer - Security Startups | Y Combinator (Freelance)

Oct 2022 - Dec 2022 | Remote

- Worked with early-stage security teams on endpoint/runtime telemetry, network observability, and detection-oriented backend services, rapidly prototyping systems from agent signals through detection APIs.
- Implemented and extended Linux telemetry flows using process and network events to reconstruct execution lineage, enrich security context, and support investigation of suspicious host behavior.
- Contributed to egress-visibility and sensitive-data monitoring concepts by correlating application/process context with outbound network activity and policy-driven detection logic.

Open Source Developer | Google Summer of Code | Apr 2022 - Sep 2022 | Remote

Transformed Node-RED toward a collaborative SaaS model; built four flow-collaboration capabilities and integrated OAuth-based authentication and authorization.

Website Developer - SPARK | Indian Institute of Technology Roorkee | Aug 2020 - Mar 2022 | Roorkee, India

Built and operated full-stack application workflows that scaled through 20,000+ annual applications, including load balancing and automated data processing.

TECHNICAL SKILLS

Security: eBPF, runtime security, process lineage, network interception, threat detection & hunting, DLP, DSPM, data lineage, network observability, access control, data masking

Languages: C++, Go, Java, Python, JavaScript

Systems & Cloud: Linux, distributed systems, microservices, AWS, GCP, Docker, Kubernetes, Terraform, CloudFormation, AWS Step Functions

Data & AI: SQL, MongoDB, Elasticsearch, ETL pipelines, NLP, LLMs, LangChain, ML-based data classification

EDUCATION

Bachelor of Technology | Indian Institute of Technology, Roorkee

Selected projects: Automating Ultrasound Diagnosis using AI; Extreme Learning Machine-based Bearing Failure Prognosis Systems